01 / EVENT & MISSION

The Straight Rows Hackathon

Where precision agriculture meets Space Domain Awareness

INDIVIDUAL	24-HOUR EVENT	NDSU CAMPUS	ONE WINNER
One participant per solution. Your code, analysis, and demonstration.	Build for 22 hours; demonstrations and judging in the final two hours.	Open to NDSU undergraduate students who are U.S. citizens.	Paid summer internship + full one-year scholarship, tuition paid.

“My tractor is drifting. Is it the Sun, an adversary, or my own hardware?”

Build software that detects degraded guidance, attributes its cause using farm telemetry and space-related feeds, and recommends an action with evidence a human operator can check.

Why this problem matters

Precision guidance helps tractors plant straight rows with minimal overlap and waste. When navigation degrades, a farmer must decide whether to stop a fleet, report interference, service one machine, or dismiss a brief reflection. A wrong diagnosis can waste a planting window or leave an interference event unexplained.

The May 2024 Gannon geomagnetic storm disrupted GPS-guided farming during planting season. Reported GPS interference affecting farms near Finland’s eastern border illustrates a different cause with similar symptoms. Straight Rows turns this real operational problem into a controlled, synthetic software challenge.

A short primer: the two domains

On the farm

GNSS means satellite navigation; GPS is one constellation. **RTK** (Real-Time Kinematic) uses corrections from a fixed base station to support centimeter-level positioning under favorable conditions.

Watch the receiver’s satellite count, **C/N0** signal strength, fix state, and correction-link status. **Cross-track error** measures deviation from an intended guidance line. Reported position can be misleading during spoofing.

Through space

Space Domain Awareness (SDA) provides context about activity and conditions affecting space systems. **Kp** and **Dst** describe geomagnetic activity; **S4** is a scintillation proxy closer to navigation-signal disturbance.

The package also includes constellation advisories and imperfect radio-frequency interference reports. Their timing and uncertainty matter. These are simulated feeds inspired by SDA data products, with no live API access required.

Participation and award. Individual work only. No prior GNSS background is required. Any programming language, internet resources, and AI coding assistants are allowed. The internship is subject to Bluestaq’s standard internship eligibility and hiring requirements. All challenge data is synthetic and unclassified.

02 / CHALLENGE & DATA

Find the cause. Choose an action.

Detect episodes, identify affected tractors, and support each decision

Analyze a simulated three-tractor fleet near Casselton, North Dakota. For each evaluation episode, report its start and end, affected machines, cause, action, confidence, and evidence. Use farm observations together with SDA context; explain when a feed supports an attribution and when it rules out an alternative.

CAUSE CLASS	WHAT TO INVESTIGATE	EXPECTED ACTION
`SPACE_WEATHER`	Regional signal degradation with elevated scintillation; Kp can precede farm impact.	`HALT_OR_DEGRADED_MODE`
`JAMMING`	Excess signal loss from RF interference. A local source can affect several machines.	`HALT_AND_REPORT`
`SPOOFING`	Position or guidance error that drifts without the expected signal-health collapse.	`HALT_CROSSCHECK_AND_REPORT`
`EQUIPMENT_FAULT`	Machine-specific antenna dropouts or a correction-link fault; compare neighboring machines.	`SERVICE_MACHINE_OTHERS_CONTINUE`
`MULTIPATH`	Brief reflections near a headland obstruction, recurring at the same location.	`CONTINUE_FALSE_ALARM`

Causes may overlap. A storm does not rule out jamming or a machine fault. Avoid assuming all machines share one cause. In this simulation, cross-track error is a separate guidance-monitor channel; reported GNSS position is not trusted ground truth.

The package you receive at kickoff

DATA	CONTENTS
`data/farm/tractor_telemetry.csv.gz`	72 hours × 3 tractors × 1 Hz = 777,600 rows. UTC time, reported position, speed, cross-track error, satellite count, mean/minimum C/N0, fix state, and correction-link status.
`data/sda/`	Three-hourly Kp, hourly Dst, 15-minute S4, constellation advisories, and delayed, incomplete RF-interference reports with location uncertainty.
`data/labels/train_episodes.json`	12 labeled development episodes. Starter notebook, baseline script, submission template, and validator are also included.

Development versus evaluation

Development: May 4, 2026 at 00:00 UTC through May 6 at 00:00 UTC (hours 0–48), with 12 labeled episodes.
Evaluation: May 6 at 00:00 UTC through May 7 at 00:00 UTC (hours 48–72), with eight unlabeled episodes. Only this window is scored.

These are **simulated dates**, not the event schedule. Machines work approximately 06:00–22:00 local time and park overnight. Scored episodes concern working tractors; poor reception while parked alone is not a scored guidance event. RFI reports may arrive 10–30 minutes late, miss events, or be false alarms. No physical equipment or RF transmission is needed.

03 / DELIVERABLES & RULES

A reproducible solution and a clear demo

Submit at hour 22; present and answer questions during the final two hours

1. Your results: submission.json

Follow `submission/submission_template.json`. Include `participant.github_handle` (your GitHub username), `repository_url`, and a non-empty `detections` list. **Submissions are public: never include your email address or student ID.** Each detection requires:

FIELD	REQUIREMENT
`start, end`	ISO 8601 timestamps with a timezone; UTC is recommended. Start must be in the evaluation window, end must follow start and be no later than May 7 at 00:00 UTC.
`tractors`	A non-empty list of distinct affected IDs: T1, T2, and/or T3.
`class, action`	Use the cause and action identifiers on page 2.
`confidence`	A number from 0 to 1. Explain uncertainty; a confident wrong answer weakens decision quality.
`evidence`	A non-empty list of strings citing timestamps, machines, measurements, and relevant SDA observations. Make the attribution checkable.

```
python submission/validate_submission.py submission.json
```

Validation checks structure, not correctness. Keep distinct concurrent causes as separate detections, and note when several belong to one operational event. The baseline is a starting point; its evidence text is intentionally incomplete.

2. Your repository

Keep your repository private until hour 22, then make it public and open one pull request to `Bluestaq/ndsu-hackathon-submissions` adding `submissions/<github-username>/submission.json`. Its opening time is your submission time. Include source code and a README with exact setup and run commands, pinned dependencies, and the command that regenerates your results. Explain your approach, how you used the labeled data, important failure modes, and what additional farm or SDA data would improve attribution. Include any required assets or clearly document how to obtain them.

3. Your demonstration

Give a **five-minute demonstration** of a working solution, followed by **three minutes of Q&A**. Show an episode, the evidence behind its cause and action, an ambiguous case or limitation, and how your code reproduces the submitted output. Judges may ask you to explain or modify your implementation.

Tools are open. The work is individual.

Any language or framework, full internet access, and AI coding assistants are allowed; no AI disclosure is required. Collaboration between competitors and sharing solutions are prohibited. You must understand and defend the work you submit.

Timing and reproducibility. The dataset is released at kickoff; code and results lock at hour 22. Record external services or credentials and provide a reproducible local path when possible. Retrospective analysis is allowed; onset accuracy does not establish real-time performance.

04 / JUDGING & LOGISTICS

How the competition is judged

100 points across execution, decisions, research, rigor, and communication

CRITERION	PTS	WHAT JUDGES ASSESS
Technical execution	30	Working detection, attribution, and recommendation system; sensible architecture; code quality; accuracy and episode timing.
Mission relevance & decision quality	25	Appropriate operator actions, calibrated confidence, and evidence that connects farm observations with SDA context. Explain ambiguity and operational consequences.
Innovation & research depth	20	Thoughtful methods and data fusion, useful improvements beyond the starter baseline, and understanding of the problem's limitations.
Assurance & rigor	15	Evaluation on development data, false-alarm and failure analysis, robustness, and results reproducible from the README.
Demo & communication	10	Clear explanation, a working demonstration, and well-supported answers to questions.

Judges average their scores. Ties are broken first by mission relevance and decision quality, then technical execution. The organizer's automated diagnostic is **supporting evidence**; it is not an additional component of the 100-point event score.

What the automated diagnostic measures

The diagnostic reports **25 × macro-F1 across the five causes + 15 × mean onset credit**, out of 40. Detections and truth episodes match one-to-one, requiring temporal intersection-over-union of at least 0.30 and affected-tractor set intersection-over-union of at least 0.50. Eligible same-class matches take priority; matching then maximizes the number of matches and total time overlap.

Onset credit is full within five minutes of the true start, declines linearly to zero at 60 minutes, and is zero for a miss or wrong class. Both early and late errors count. Extra detections reduce precision. Recurrent multipath reports lasting at most three minutes are exempt from extra-detection penalties except duplicates overlapping the scored representative window; credit still requires matching that representative. Judges also review whether the reports and their evidence make operational sense.

Registration, schedule, and award

The event takes place on the **NDSU campus**. Registered participants receive the confirmed date, room, and kickoff time from Bluestaq University Programs. The agenda, step-by-step instructions, opening remarks, and results are at bluestaq.github.io/ndsu-hackathon-submissions. For registration and questions, contact universityprograms@bluestaq.com.

One winner receives **both a paid summer internship at Bluestaq headquarters in Colorado Springs and a full one-year scholarship that pays tuition**. The internship is subject to standard internship eligibility and hiring requirements. Participation is restricted to NDSU undergraduate students who are U.S. citizens.

Starter reading and public resources

May 2024 storm and tractor GPS disruption — 404 Media

GPS interference affecting Finnish farms — Future Farming

Ionospheric scintillation in precision agriculture — GPS World

Recent Advances on Jamming and Spoofing Detection in GNSS — Sensors (2024)

NOAA SWPC

GPS status advisories

GPS interference maps

Unified Data Library